

Observer-Patch Holography and the Dark Matter Phenomenon

The Modular-Anomaly Source

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Abstract

In observer-patch holography the Einstein equation sources curvature from the modular-charge stress of the screen under a universal coupling. Every screen source that carries modular charge therefore gravitates, whether or not it carries the electromagnetic readout, and the luminous-only Einstein relation holds exactly when the non-luminous charge vanishes. A dark sector is generic. The candidate source is the anomalous modular energy carried on overlap collars, the same finite-stage remainder that the Einstein derivation controls. Its rest-frame stress is bounded by the collar recovery defect, vanishes at full recovery, and dilutes as the inverse cube of the scale factor when conserved. These statements are machine-checked. The source must meet two data boundaries: on the SPARC rotation-curve sample the deep-regime law reproduces the radial-acceleration and Tully–Fisher relations with one constant near $1.4 \times 10^{-10} \text{ ms}^{-2}$, and the Cassini quadrupole excludes any universal fixed-coefficient extension of that law at nineteen standard deviations. The magnitude of the anomalous charge, its radial profile, and the vanishing of the recovery defect at high record density are work in progress.

Claim Boundary

The finite OPH results used here are the rest-frame Einstein relation on its named small-ball premises, the decomposition of the cap modular generator into a bulk part, a central part, and a collar-carried anomalous part, and the controlled bound on that anomalous part [1, 2]. The theorems of Sections 2 and 3 are algebraic consequences of those objects and are checked in the Lean formalization [3]. Universal coupling is a premise of the Einstein branch, named as a structure field in the formalization and not established there.

What the paper does not supply: a quotient-invariant receipt localizing the anomalous modular energy on collars, the value of the collar constant, the radial profile of the anomalous charge around a baryonic source, a proof that the recovery defect vanishes in any physical regime, and any cosmological likelihood. No cosmological result carries an OPH falsification verdict.

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1 What Sources Gravity in OPH

The Einstein arrow of the spacetime paper ends in

$$G_{ab} + \Lambda g_{ab} = 8\pi G \langle T_{ab} \rangle, \quad (1)$$

where $\langle T_{ab} \rangle$ is the local conserved stress reconstructed from positive null translations and modular charges of the record algebra [1]. In the rest frame the relation is obtained from three premises on a small geodesic ball of radius ℓ : generalized entropy is stationary, the bulk entropy variation equals $(8\pi^2 \ell^4/15) \delta \langle T_{00} \rangle$, and the fixed-volume area variation equals $-(4\pi \ell^4/15) \delta G_{00}$. Exact arithmetic then gives $\delta G_{00} = 8\pi G \delta \langle T_{00} \rangle$.

The stress entering this relation is modular charge. The screen does not consult the electromagnetic readout of a source before responding to it. That is the content of the universal-coupling premise, and it is the only input the dark-sector question needs from the gravity derivation.

2 The Generic Dark Sector

Let a finite family of screen sources be indexed by i , each with rest-frame modular-charge contribution q_i and a flag recording whether it carries the electromagnetic readout. Write Q_{lum} for the sum over flagged sources and Q_{dark} for the sum over the others, so that $Q_{\text{tot}} = Q_{\text{lum}} + Q_{\text{dark}}$.

Definition 2.1 (Universal coupling). *The stress contraction entering the rest-frame Einstein relation equals Q_{tot} , with no readout-dependent weight.*

Theorem 2.2 (Geometry responds to all modular charge). *Under universal coupling, $\delta G_{00} = 8\pi G (Q_{\text{lum}} + Q_{\text{dark}})$.*

Theorem 2.3 (The luminous-only relation). *Under universal coupling, $\delta G_{00} = 8\pi G Q_{\text{lum}}$ holds if and only if $Q_{\text{dark}} = 0$.*

Corollary 2.4 (A geometric excess forces a dark source). *If $\delta G_{00} \neq 8\pi G Q_{\text{lum}}$, some source without the electromagnetic readout carries nonzero modular charge.*

Proposition 2.5 (Attractive sign). *If every non-luminous charge is nonnegative, $Q_{\text{dark}} \geq 0$, and the dark contribution enters the weak-field equation with the sign of baryonic mass.*

The proofs are finite sums and the positivity of $8\pi G$. Their value is the direction of the inference. In a theory where gravity is sourced by modular charge, the question is not whether a dark sector exists; any geometric excess over the luminous prediction is by itself the statement that non-luminous modular charge is present. The open questions are which screen object carries that charge, how much of it there is, and how it is distributed.

3 The Modular-Anomaly Source

On the support-visible branch the cap modular generator has the local form

$$K_C = 2\pi B_C + K_C^{\text{anom}} + \text{const}, \quad (2)$$

with B_C the bulk boost generator and K_C^{anom} the nonadditive part carried on the overlap collar [1]. At finite regulator stage the Einstein derivation carries this part as a remainder and bounds it:

$$|\delta\langle E_C^{(\delta)} \rangle| \leq C_C \eta_\delta, \quad (3)$$

where η_δ is the collar recovery defect and C_C depends only on the collar model and the bounded observable class.

Definition 3.1 (Anomalous stress).

$$\delta\langle T_{00}^{\text{anom}} \rangle := \frac{15}{8\pi^2 \ell^4} \delta\langle E_C^{(\delta)} \rangle. \quad (4)$$

This is the Einstein branch's own bookkeeping term. The dark-sector proposal is that it is not bookkeeping: the anomalous modular energy is a physical non-luminous charge in the sense of Section 2. Three consequences follow from (3) and (4) alone.

Theorem 3.2 (Bounded source). $|\delta\langle T_{00}^{\text{anom}} \rangle| \leq 15 C_C \eta_\delta / (8\pi^2 \ell^4)$.

Theorem 3.3 (Full recovery switches the source off). *If $\eta_\delta = 0$ then $\delta\langle T_{00}^{\text{anom}} \rangle = 0$. Along any family with fixed ℓ , bounded C_C , and $\eta_\delta \rightarrow 0$, the anomalous stress tends to zero.*

Theorem 3.4 (Cold scaling). *A conserved anomalous charge in a comoving cell of physical volume $\propto a^3$ has density $\propto a^{-3}$.*

Theorem 3.3 is the screening interface. Wherever repair saturates and the collar recovery defect vanishes, the dark source is absent. Whether η_δ vanishes in the high-acceleration, high-record-density regime of a planetary system is the physical question on which the Solar-System bound below turns; the theorem states what follows if it does.

4 Data Boundaries the Source Must Meet

4.1 Deep regime

A static spherical source law of the form

$$a_{\text{dark}} = \sqrt{a_b a_0}, \quad v^4 = G M_b a_0, \quad (5)$$

is the deep-regime target. One OPH mechanism for its exponent is a codimension-one collar count: repair opportunities on closed cuts around a point source scale as $\sqrt{g_b/a_0} = r_M/r$ rather than as the area count g_b/a_0 [2]. That count is a premise, not a consequence of Section 3.

The law is compared with the SPARC sample of 175 disk galaxies with $3.6\mu\text{m}$ photometry [5] under the standard cuts of McGaugh, Lelli, and Schombert [6]: quality flag at most 2, inclination at least 30° , velocity error at most ten percent, stellar mass-to-light ratios 0.5 for disks and 0.7 for bulges, leaving 2700 points in 149 galaxies [4]. On the self-consistent deep subset $g_{\text{bar}} < fa_0$ the fixed-exponent fit gives

$$a_0 = 1.57 \times 10^{-10}, \quad 1.40 \times 10^{-10}, \quad 1.28 \times 10^{-10} \text{ m s}^{-2}$$

for $f = 0.3, 0.1, 0.03$, with 1651, 1043, 233 points and rms scatter between 0.14 and 0.17 dex; the free exponents are $0.60 \pm 0.01, 0.57 \pm 0.02, 0.39 \pm 0.08$. For the 123 galaxies with a measured flat velocity, $M_b = 0.5L_{[3.6]} + 1.33M_{\text{HI}}$, the fixed-exponent Tully–Fisher normalization gives $a_0 = 1.55 \times 10^{-10} \text{ m s}^{-2}$ with 0.26 dex scatter and free exponent 3.75 ± 0.10 . The two constants differ by 0.046 dex, inside the combined scatter. The exponents are the parameter-free content of (5); a_0 is a fitted comparison value, and no OPH construction supplies it.

4.2 Solar System

Applied unchanged to the Sun inside the measured Galactic field, the collar-count law with the fixed coefficient $1 - P/24$ predicts an external-field quadrupole $Q_2 = 3.62 \times 10^{-26} \text{ s}^{-2}$, against the Cassini value $(1.6 \pm 1.8) \times 10^{-27} \text{ s}^{-2}$ [7]: a 19σ exclusion at fixed inputs, and about 11σ after the Gaia uncertainty on the Galactic acceleration [8]. Equivalently, the additive form $g = g_{\text{bar}} + \sqrt{g_{\text{bar}}a_0}$ predicts a relative anomaly of 1.5×10^{-4} at one astronomical unit against a bound near 10^{-8} . Any universal fixed-coefficient extension of (5) is excluded.

This is why Theorem 3.3 is the first open target rather than a remark. A suppression rule introduced after reading Cassini would be a fitted continuation. A proof that the collar recovery defect vanishes where record density saturates would be the same object that sources the halo switching itself off, and it would be decided before any further data are examined.

5 Open Targets

1. **Screening from recovery.** Prove or refute that $\eta_\delta \rightarrow 0$ in the repair-saturated regime of a compact source, with a rate in the local acceleration or record density. Theorem 3.3 then gives the Solar-System suppression; a refutation retires the modular-anomaly source.
2. **Magnitude.** Attach the normalization of $\delta\langle E_C^{(\delta)} \rangle$ to repair conservation on the Ward/Bianchi rung of the gravity premise ladder, so that a_0 is a consequence rather than a value borrowed from H_0 and Λ .
3. **Profile.** Derive the codimension-one collar count, or replace it, from the collar geometry of a point source, so that the exponent in (5) is a theorem on the same premises as Section 3.
4. **Localization.** Supply the quotient-invariant receipt identifying the collar operator in (2) with the small-ball remainder in (4).

A rotor completion, with integer repair occupation and a conjugate phase, is admissible only under the identification of the occupation with the anomalous modular charge and of the phase with the modular flow angle. Without that identification it is a parametrization of items 2 and 3 rather than a derivation, and it is not used here.

6 Conclusion

Under universal coupling the screen gravitates according to its total modular charge. A dark sector is therefore a theorem-level feature of OPH, and any geometric excess over the luminous prediction is the presence of non-luminous modular charge. The anomalous modular energy on overlap collars is the one candidate for that charge that the Einstein derivation itself carries; it is bounded by the recovery defect, absent at full recovery, and cold when conserved. The deep-regime galaxy data fix the target its profile must reach, and the Cassini bound fixes the target its screening must reach. The proofs that close those targets are work in progress.

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